

sible scenarios have been covered, and training lasted for a reasonable duration.

- AI systems are thoroughly tested before deployment, and we keep checking them regularly.

- Things surrounding the AI system are also well integrated with it because, sometimes, the AI system may be good enough, but other interactive systems may throw it off.

By covering these three aspects, I believe we can have reasonable level of trust in an AI system.

Governing automated decision systems

What about the automated decision systems, especially the ones that may have a profound impact on social or personal life? How can we oversee such systems?

I think the best way to implement governance for any automated decision-making system is by making it follow the fundamental rules of the law.

The rule of law says that decision-makers must be transparent and accountable—make an AI system transparent and fix accountability.

The rule of law also says that the decision-maker must be predictable and consistent—make sure that the AI system is predictable and consistent, too.

And finally, the rule of law also says that as a decision-maker, you must always put equality before the law—so, the AI system must be free of bias and discrimination, even if it means diverging from some of the set rules in that case.

In general, I strongly feel the need to have a responsible human being always in the loop when crucial decision making is required. An automated decision system should always recommend not to decide.

Responsible AI practices

There are five practices that each company can follow to ensure the responsible use of AI systems.

1. Keeping a balanced scorecard for AI systems' training and monitoring or say governance. When you have several metrics in a stable scorecard form, you can easily understand how they interact with each other.

How maximising or minimising one metric can affect other parameters, and so on.

2. Keeping humans in the loop, always. Technology systems may know how to achieve a goal, but they lack common sense, they lack human-level reasoning. So, unless you have a responsible adult who is in charge of this juvenile AI system, there are many ways it could go wrong.

3. Keeping a close eye on inputs. There is a reason why we say—*garbage in garbage out*. If your data has flaws, your AI system will manifest them in output, but at scale. Minor flaws would get amplified to cause problems. There are a few characteristics of a unique full-spectrum dataset—it is timely, complete, correct, integral, accurate, appropriate balance, and it is sufficient. So ensuring that full-spectrum data is available is the key.

4. Knowing its limits. AI systems are not all in one solution. Even narrow AI solutions have limitations; they have asymptotic points, that is, situations where the system fails to function. By understanding those limits, everyone can avoid them or manage them effectively.

5. Testing thoroughly and keep doing it regularly. First of all, you should always check your AI system for all the potential scenarios. Even if they are only remotely possible or are long-tail cases—testing the system for those cases will ensure that everything is in order.

I often talk about 'Concept Drift.' It is the phenomenon where the machine learning model characteristics can change over some time. It means, if you trained and built your system by referring to a particular dataset, it may no longer be relevant. With the change in time and several external factors, the model may be invalid or suboptimal. It is the best practice to test your AI system for concept drift, regularly.

Business challenges in ensuring the responsible use of AI

Many of the businesses are confused as to what to do when it comes to Responsible AI. Simply because there are too many people out there advising companies with Responsible

AI and Ethical AI frameworks. The problem with most of them is a conflict of interest. When there is such a conflict, i.e., you are advising and providing AI solutions, too, it rarely yields positive results.

At the same time, many businesses are not even there yet. They are merely ensuring that best practices are followed; they are using standard protocols or frameworks and are maintaining project management hygiene. The issue is that people haven't understood the vastness, and the scope of the risk when AI goes wrong.

The fact that we don't know what we don't know is a significant impediment. It is restricting companies from visualising this problem.

Unlocking the full potential of the workforce

While we take care of risks and other issues related to Responsible AI, how do we unlock the full potential of the workforce?

It is a relatively sensitive issue. There is constant unrest amongst the workforce in several companies. The concern of losing jobs is real and challenging.

However, first, I would recommend that companies should use the AI system only in places where the workforce is complaining about hard work. Whatever your workforce is happily doing, why touch it?

Second, you can use AI solutions, where the time is an essence, and adding more members to the workforce is not a solution. Millisecond decisions or similar ones are good candidates for automation with AI.

Third, use AI solutions where unpredictability is costing more or not bringing more revenue. For example, use it for top-line improvement. If you focus on bottom-line improvement, it will only give measurable efficiency gains.

And for everywhere else, I don't see an immediate need for an AI solution.

Several years ago, I was working with a company on a time-saving project and even after three months in, we did not achieve any headway. I wasn't clear what were we missing until one day, one of the colleagues